

Lab 4: Creating, Viewing, and Editing Text Files & Using Shell Redirection, Pipes, and Variables in RHEL 9

Section 1: Save Command Output or Errors to a File with Shell Redirection

1. **Question:** Use the `ls` command to list the contents of the `/etc` directory and redirect the output to a file named `etc_list.txt`.

Answer:

```
abdelhamed@eng-abdelhamed:~/Tests$ ls -l &>> etc_list.txt
abdelhamed@eng-abdelhamed:~/Tests$ ls -l &>> etc_list.txt
abdelhamed@eng-abdelhamed:~/Tests$ cat etc_list.txt
total 0
-rw-r--r-- 1 abdelhamed abdelhamed 0 Dec  6 18:29 etc_list.txt
ls: cannot access 's': No such file or directory
ls: cannot access 's': No such file or directory
total 4
-rw-r--r-- 1 abdelhamed abdelhamed 169 Dec  6 18:29 etc_list.txt
ls: cannot access 's': No such file or directory
```

2. **Question:** Run the `cat` command on a non-existent file (`nofile.txt`) and redirect the **error output** to a file named `error_output.txt`.

Answer:

```
abdelhamed@eng-abdelhamed:~/Tests$ cat nofile.txt 2>> error_output.txt
abdelhamed@eng-abdelhamed:~/Tests$ cat error_output.txt
cat: nofile.txt: No such file or directory
abdelhamed@eng-abdelhamed:~/Tests$
```

Question: Combine the output and error from the find command into a single file named find_results.txt, searching for any files named *.log in the /var/log directory.

3. Answer :

```
abdelhamed@eng-abdelhamed:~/Tests$ find /var/log -iname "*.log" &>> find_result.txt
abdelhamed@eng-abdelhamed:~/Tests$ cat find_result.txt
/var/log/apt/term.log
/var/log/apt/history.log
/var/log/bootstrap.log
find: '/var/log/private': Permission denied
/var/log/dpkg.log
/var/log/alternatives.log
/var/log/upgrade-policy-changed.log
/var/log/ubuntu-advantage.log
abdelhamed@eng-abdelhamed:~/Tests$
```

Section 2: Process Command Output Through Multiple Command-Line Programs Using Pipes

4. **Question:** List all files in the /var/log directory, and pipe the output to the grep command to filter only for files containing "auth" in their name.

answer: `ls /var/log | grep -l "auth"`

5. **Question:** Run the ps -aux command to list running processes, then pipe the output through the sort command to sort by CPU usage

```
abdelhamed@eng-abdelhamed:~/Tests$ ps -aux | sort -k3 -nr
root      10  0.2  0.0   2624   132 ?        S      18:28   0:02 /init
root      9   0.0  0.0   2624   124 ?        Ss     18:28   0:00 /init
root      5   0.0  0.0   2616     4 hvc0     Sl+    18:28   0:00 plan9 --c
root      1   0.0  0.0   2616  1760 hvc0     Sl+    18:28   0:00 /init
abdelha+ 223  0.0  0.0  19800  1164 pts/0    S+     18:45   0:00 sort -k3
abdelha+ 222  0.0  0.0   7796  3088 pts/0    R+     18:45   0:00 ps -aux
abdelha+ 150  0.0  0.0   7808  3604 pts/0    T      18:41   0:00 top
abdelha+ 11   0.0  0.0   6348  5452 pts/0    Ss     18:28   0:00 -bash
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
abdelhamed@eng-abdelhamed:~/Tests$
```

6. **Question:** Display all lines of a file named example.txt in uppercase using a combination of cat and tr (translate).

Answer :

```
abdelhamed@eng-abdelhamed:~/Tests$ cat find_result.txt | tr [:lower:] [:upper:]
/Var/log/apt/term.log
/Var/log/apt/history.log
/Var/log/bootstrp.log
FIND: '/Var/log/private': PERMISSION DENIED
/Var/log/dpkg.log
/Var/log/alternatives.log
/Var/log/upgrade-policy-changed.log
/Var/log/ubuntu-advantage.log
abdelhamed@eng-abdelhamed:~/Tests$
```

Section 3: Create and Edit Text Files Using the Vim Editor

7. **Question:** Create a new file named notes.txt using the vim editor. Add the following text to the file and save it:

text

Copy code

Linux is a powerful operating system.

It is widely used for servers and development.

```
ubuntu:~$ vim dasd
```

```
ubuntu:~$ cat dasd
```

text

Copy code

Linux is a powerful operating system.

It is widely used for servers and development.

```
abdelhamed@eng-abdelhamed:~/Tests$ vim notes.txt
abdelhamed@eng-abdelhamed:~/Tests$ cat notes.txt
text
copy code
Linux is a powerfil opertating system
it is widely used for servers and development
abdelhamed@eng-abdelhamed:~/Tests$
```

8. **Question:** Open the notes.txt file again in vim, go to the second line, and append the text: It is open-source and secure. Save the file and exit.

Answer: First vim notes.txt

Next: 2G this refers to second line

Third: A -> For Append, then write It is open-source and secure , then escape , :wq write and quite

9. **Bonus Question:** In vim, search for the word "Linux" in the notes.txt file and replace it with "GNU/Linux"

10. Answer:

```
~
:%s /Linux/GNULinux/g
```

Section 4: Use Shell Variables to Help Run Commands

10. **Question:** Create a shell variable named MY_VAR and assign it the value "Hello, World!". Display the value of the variable.

```
abdelhamed@eng-abdelhamed:~/Tests$ export MY_VAR="Hello World!"
abdelhamed@eng-abdelhamed:~/Tests$ echo $MY_VAR
Hello World!
```

11. **Question:** Run the command `echo $HOME` to display your home directory. Now assign the output of this command to a variable named `MY_HOME`, and use `echo` to display the value of `MY_HOME`.

```
abdelhamed@eng-abdelhamed:~/Tests$ MY_HOME=$(echo $HOME) ; echo $MY_HOME
/home/abdelhamed
abdelhamed@eng-abdelhamed:~/Tests$
```

- 12.
13. **Bonus Question:** Use a variable named `DIR_NAME` to dynamically change the directory when running the `cd` command.

Answer: `DIR_NAME=/etc ; cd $DIR_NAME`

Section 5: Edit Bash Startup Scripts to Set Shell and Environment Variables

13. **Question:** Open your `.bashrc` file in `vim` and add a line that sets an environment variable `EDITOR` to `vim`. Save the file and restart your terminal.

14.

```
abdelhamed@eng-abdelhamed:~$ echo $EDITOR
vim
abdelhamed@eng-abdelhamed:~$
```

15.

Question: Add a new alias to your `.bashrc` file that allows you to run the `ls -la` command using just `ll`. Save the file and reload your shell.

16. Answer:

```
abdelhamed@eng-abdelhamed:~$ abdelhamed
total 72
drwxr-x--- 6 abdelhamed abdelhamed 4096 Dec  6 19:23 .
drwxr-xr-x 3 root      root      4096 Apr 20  2024 ..
-rw----- 1 abdelhamed abdelhamed 7667 Dec  6 18:15 .bash_history
-rw-r--r-- 1 abdelhamed abdelhamed  220 Apr 20  2024 .bash_logout
-rw-r--r-- 1 abdelhamed abdelhamed 3816 Dec  6 19:23 .bashrc
-rw-r--r-- 1 abdelhamed abdelhamed   79 Apr 26  2024 .c
drwx----- 4 abdelhamed abdelhamed 4096 Dec  6 18:41 .config
-rw----- 1 abdelhamed abdelhamed   20 Dec  6 19:06 .lessht
```

17. **Bonus Question:** Set a custom `PATH` by appending the directory `/usr/local/bin` to your current `PATH` variable in your `.bash` profile. Save and verify the change.

18. `export PATH="$PATH:/usr/local/bin"`

In `.bashrc`